# Danesh Morales Hashemi

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Updated on October 30, 2026

## Research Positions

**2025 – Present** **Graduate Research Assistant**, Institute for Quantum Computing
Advisors: Graeme Smith and Ningping Cao

**2024 – 2025** **Undergraduate Research Assistant**, Institute for Quantum Computing
Advisor: Joseph Emerson
Topic: *Circuit Benchmarking for Efficient and Scalable Fidelity Estimation*

**Fall 2023** **Undergraduate Research Assistant**, University of Waterloo
Advisor: Roberto Guglielmi
Topic: *Controllability of Fokker-Planck Equation*

## Education

**2025 – Present** **MMath in Applied Mathematics**, University of Waterloo
Quantum Information Specialization

**2020 – 2025** **BMath in Mathematical Physics**, University of Waterloo
Co-op Program • Pure Mathematics Minor • Graduated with Dean's Honours

## Publications

1. **What Must a Quantum-Memory Decoder Know About Temporally Correlated Noise?**
   **Danesh Morales-Hashemi**, Graeme Smith, and Ningping Cao.
   Preprint, 2026 • arXiv:2610.03545 [quant-ph]

## Conferences and Talks

1. **Demonstrating Circuit Benchmarking for Scalable and Efficient Fidelity Estimation**
   **APS March Meeting 2025**
   Session on Scalable Verification and Validation (QCVV)
   Anaheim Convention Center • March 2025

## Teaching Experience

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|---|---|
| Spring 2025 | **Teaching Assistant**, AMATH 250 – Introduction to Differential Equations |
| Fall 2023 | **Instructional Support Assistant**, MATH 115 – Linear Algebra for Engineering |
| Winter 2022 | **Instructional Support Assistant**, CS 115 – Introduction to Computer Science |

## Projects

1. **Introduction to Cycle Benchmarking for Quantum Error Diagnostics**
   QIC 820 Final Project, Institute for Quantum Computing • December 2025
2. **Predicting MLB Playoffs Using Quantum Machine Learning**
   QIC 890 Final Project, Institute for Quantum Computing • December 2025
   GitHub repository
3. **Demonstrating Circuit Benchmarking for Efficient and Scalable Fidelity Estimation**
   Undergraduate Thesis, University of Waterloo • April 2025
4. **Introduction to Superoperator Representations and Characterization of Completely Positive Maps**
   AMATH 900 Final Project, University of Waterloo • August 2024

## Awards and Scholarships

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|---|---|
| Institute for Quantum Computing Undergraduate Research Award 2024 | **$16,000 CAD** |
| University of Waterloo President's Scholarship 2020 | **$2,500 CAD** |
| Full Undergraduate Educational Excellence Scholarship 2019, SENACYT, Panama | **$250,000 USD** |

## Relevant Coursework

(with links to syllabus)

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|---|---|---|
| Quantum Physics | Quantum Theory 1/2/3 | Representation Theory |
| Quantum Information Processing | Quantum Machine Learning | Groups and Ring Theory |
| Quantum Devices | Theory of Quantum Information | Galois and Field Theory |
| Neural Networks | Functional Analysis | Differential Geometry |